Fraction Division: How Many Groups?

Explore & Reflect

Name _____ Date _____ Period _____

Open **Fraction Division: How Many Groups?** in your unit's Interactive Math section on Mr. K's Classroom (or use your teacher's link). Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3. Do Parts 4 and 5 on your own.

1 Predict before you touch the tool

1. Is $3 \div \frac{1}{2}$ more or less than 3? Explain.

2. In $2 \div \frac{3}{4}$, two full groups of $\frac{3}{4}$ fit and $\frac{1}{2}$ of a whole is left. Is the answer $2\frac{1}{2}$? Explain.

2 Explore use the tool

1. Build $3 \div \frac{1}{2}$. How many groups fit? How many small pieces are in 3, and in one group?

2. Build $2 \div \frac{3}{4}$. How many full groups fit, and what fraction of a group is left?

3. Build $\frac{1}{2} \div \frac{3}{4}$. Why is the answer less than 1?

4. Build $3 \div \frac{3}{2}$. Then compute $3 \times \frac{2}{3}$. What do you notice?

3 Record use the tool and fill in the table

Count the small pieces

Read the pieces from the sentence under the model.

Division	Pieces you have	Pieces in one group	Answer	Same as multiplying
$2 \div \frac{1}{4}$				
$3 \div \frac{3}{4}$				
$\frac{3}{4} \div \frac{1}{2}$				
$4 \div \frac{5}{2}$				

4 Explain what did you discover?

1. $a \div d$ asks: how many _____ of size d fit into a?
2. When the divisor is less than 1, the quotient is _____ than the dividend.
3. Dividing by $\frac{n}{m}$ gives the same answer as multiplying by _____.
4. A leftover piece is a fraction of one _____, not of one whole.
5. Use the bar model to explain why $2 \div \frac{3}{4}$ is $2\frac{2}{3}$ and not $2\frac{1}{2}$.

5 Apply a new situation, no tool

1. How many $\frac{2}{3}$-cup servings are in 4 cups of rice?
2. A ribbon $\frac{7}{8}$ m long is cut into pieces $\frac{1}{4}$ m long. How many full pieces, and what fraction of a piece is left?
3. Compute $1\frac{1}{2} \div \frac{3}{8}$.
4. Without computing, is $5 \div \frac{9}{10}$ greater or less than 5? Why?